

DINESH LAMICHHANE

dineshlamxne@gmail.com | +1 (938) 248-1784 | [linkedin.com/in/dinesh-lamichhane](https://www.linkedin.com/in/dinesh-lamichhane) | lamichhanedinesh.com.np

PROFESSIONAL SUMMARY

GIS and Remote Sensing professional with 4+ years of experience in geospatial analysis, spatial data management, remote sensing, and GIS mapping. Proficient in ArcGIS Pro, QGIS, Python, SQL, and UAV photogrammetry, with expertise in spatial analysis, geoprocessing, geodatabase management, and map production. Experienced in delivering geospatial solutions for government, transportation, environmental, and disaster risk management projects through accurate spatial analysis and data-driven decision-making.

EDUCATION

[Jacksonville State University](#)

Master of Science in GIS & Technology

08/2025 - Present

Jacksonville, Alabama

[IOE Pashchimanchal Campus, Tribhuvan University](#)

Bachelor of Engineering in Geomatics Engineering

11/2017 - 07/2022

Pokhara, Nepal

PROFESSIONAL EXPERIENCE

GIS Intern | [Indiana Department of Transportation \(INDOT\)](#)

05/2026 - Present

- Supported transportation planning, roadway safety, and infrastructure projects through GIS-based spatial analysis, mapping, and geospatial data management.
- Developed GIS models, automation tools, and geoprocessing workflows to improve data analysis efficiency and streamline repetitive tasks.
- Managed and maintained transportation datasets with GIS QA/QC procedures while delivering accurate spatial products and decision-support solutions for engineering teams.

Graduate Teaching Assistant | [Jacksonville State University](#)

08/2025 - 04/2026

- Supported undergraduate and graduate GIS laboratories by mentoring more than 20 students each semester in ArcGIS Pro, spatial analysis, cartography, and geospatial data management.
- Assisted faculty with GIS research, grading, laboratory development, and curriculum improvement.
- Provided technical guidance on geoprocessing workflows, map production, spatial modeling, and Python-based GIS automation.

GIS Analyst | [NAXA](#)

03/2024 - 07/2025

- Delivered GIS solutions for 10+ government, humanitarian, infrastructure, and environmental projects through thematic maps, dashboards, StoryMaps, and spatial analysis.
- Processed UAV and satellite imagery using ArcGIS Pro, Pix4D, Agisoft Metashape, and QGIS, optimizing workflows to improve processing efficiency by ~30%.
- Conducted flood risk assessments, terrain modeling, watershed analysis, and land use mapping to support infrastructure planning and disaster risk reduction.
- Designed and maintained geospatial databases with comprehensive QA/QC procedures while collaborating with multidisciplinary teams to deliver accurate geospatial products.

TECHNICAL SKILLS

GIS & Enterprise Systems: ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, QGIS, ArcGIS StoryMaps, Dashboards, Experience Builder, Survey123, Field Maps, QuickCapture

Spatial Analysis: Network Analysis, Spatial Statistics, Terrain Analysis, Hydrologic Modeling, Geoprocessing, Cartography, Spatial Modeling, Geodatabase Management, QA/QC

UAV & Drone Mapping: Pix4D, Drone2Map, Agisoft Metashape, DroneDeploy, UAV Photogrammetry, Orthomosaic & 3D Model Generation

Remote Sensing: Google Earth Engine, Satellite Image Processing, Land Cover Classification, Change Detection, DEM/DSM, LiDAR Fundamentals, NDV, Orthomosaic Generation

Programming: Python, ArcPy, SQL, R, ModelBuilder, JavaScript (Leaflet), AI

Languages: English (Fluent), Nepali (Native), Hindi (Fluent)

SELECTED PROJECTS

(Explore all projects at: www.lamichhanedinesh.com.np/projects)

- **Traffic Signal Warrant Evaluation and GIS Automation – INDOT Vincennes District**
Created a GIS-based traffic signal warrant analysis tool following MUTCD guidelines to evaluate intersections for potential signal installation. The tool supports both two-way and one-way intersection analysis and evaluates Warrant 1, 2, and 3 criteria for intersection screening. Integrated Google Maps Street View connectivity for location verification and developed a reusable workflow that enables signal analysis for intersections across the United States.
- **Trauma Center Proximity Mapping – Indiana**
Performed network-based service area analysis in ArcGIS Pro to identify 45-minute travel-time coverage areas for trauma centers across Indiana. Created statewide accessibility maps and an interactive GIS dashboard to support healthcare accessibility analysis, project prioritization, and emergency response planning.
- **Crash Data Analysis and Heat Map Development – Vincennes District (INDOT)**
Processed and cleaned crash datasets by validating spatial coordinates and preparing GIS-ready layers. Conducted spatial analysis in ArcGIS Pro to generate district and county-level crash heat maps, identify high-risk areas, and support transportation safety planning.
- **Horizontal Curve Detection and Roadway Sign Analysis – INDOT Vincennes District**
Developed an automated ArcGIS Pro toolbox to detect horizontal curves using roadway centerline geometry. Configured parameters for interval-based bearing angle calculations, Phi thresholding, minimum curve length filtering, and curve-type classification. Performed data cleaning on segmented road layers to ensure accurate curve extraction. Integrated sign inventory data and conducted spatial analysis to determine sign presence for each curve segment, supporting roadway safety assessments and engineering decision-making.
- **Drones-Based Aerial Survey For Rapid Damage Assessment: Western Nepal Earthquake 2023**
Led drone survey operations in earthquake-affected areas of western Nepal, overseeing field mobilization, flight planning, and image acquisition. Processed high-resolution drone imagery using Pix4D, Agisoft, and DroneDeploy to generate orthomosaics and 3D models. Conducted damage assessments in ArcGIS Pro and integrated results with existing GIS datasets to produce detailed spatial reports and StoryMaps, delivering actionable insights for post-disaster recovery and decision-making.

CERTIFICATIONS & TRAINING

Foundation of GNSS | GIS for Climate Action (Esri) | Cartography (Esri) | Going Places with Spatial Analysis (Esri) | Sediment Transport with HEC-RAS | Transform AEC Projects with GIS and BIM (Esri) | Geospatial Data Analysis with ChatGPT & Google Earth Engine | Introduction to Remote Sensing | Machine Learning | Python for Data Analysis

AWARDS & ACHIEVEMENTS

- [Esri StoryMaps Competition 2024](#) - Winner, Humanitarian and Disaster Response Category (International competition)
- Outstanding Student Award in GIS & Technology - Jacksonville State University, 2025

REFERENCES

Dr. Saeideh Gharehchahi

Assistant Professor of Geography
Jacksonville State University, Alabama, USA
Contact: +1 (256) 365-4414
E-mail: sgharehchahi@jsu.edu

Er. David Reamer

Highway Engineer Supv 4
Indiana Department of Transportation, Indiana, USA
Contact: +1 (812) 830-9979
E-mail: dreamer2@indot.in.gov